MTL HACKATHON · DEVICE TRACK

Chilly Chiplet

Chip layout design GUI
That allows real-time thermal eval

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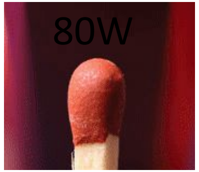
Importance

Heat on Chips is an issue!

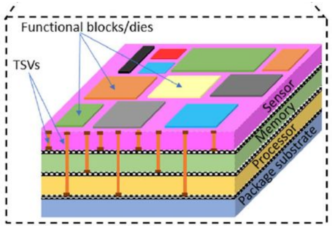


Next-Generation GPU-HBM Roadmap
: More GPU & HBM Integrated Above Interposer

GPU Architecture	Rubin (2026)	Feynman (2029)	Post Feynman (2032)	Next-Gen Architecture (2035)
GPU Die Size	728 mm²	750 mm²	700 mm²	600 mm²
GPU Power	800 W	900 W	1,000 W	1,200 W
GPU-HBM Module	R200	F400	Post Feynman GPU-HBM Module	Next-Gen GPU-HBM Module
Interposer Size (GPU, HBM, Interposer)	47.5 mm × 46.2 mm	56.2 mm × 85.2 mm	58.5 mm × 102.8 mm	95.9 mm × 96.4 mm
# of GPU Dies	×2	×4	×4	×8
# of HBM Stack	HBM4×8	HBM5×8	HBM6×16	HBM7×32
Interposer Die Size	2,194 mm² (46.2 mm x 48.5 mm)	4,788 mm² (85.2 mm x 56.2 mm)	6,014 mm² (102.8 mm x 58.5 mm)	9,245 mm² (96.4 mm x 95.9 mm)
Total Bandwidth	16 / 32 TB/s	48 TB/s	128/256 TB/s	1,024 TB/s
Total HBM Capacity	288/384 GB	400/500 GB	1,536/1,920 GB	5,120/6,144 GB
Total Power	2,200 W	4,400 W	5,920 W	15,360 W

HBM Roadmap Ver 1.7 Workshop by KAIST TERALAB

Compression to 2.5D/3D to increase yield/modularity exacerbate thermal issues!



3-D heterogeneous integration

Sharma & Ramos-Alvarado (2026)

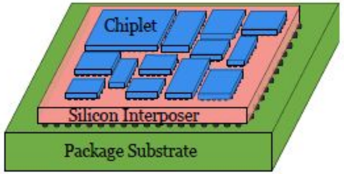
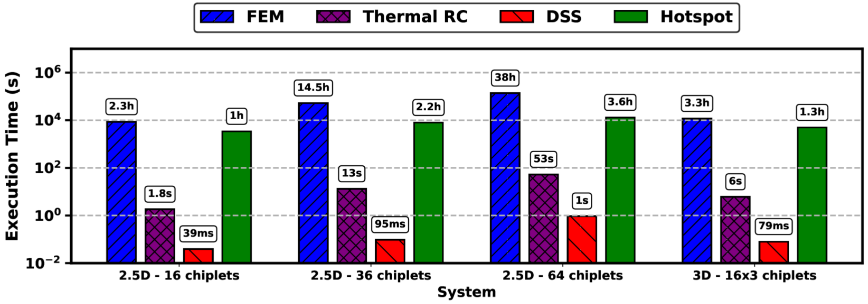


Fig. 2 Floorplan design of chiplets in 2.5D package.

Chen, et al (2023)

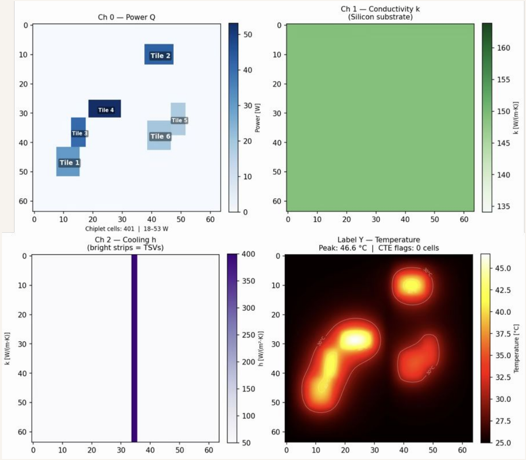
Time (2-30 hours each trial) for iterative FEM simulation to identify thermal issues!



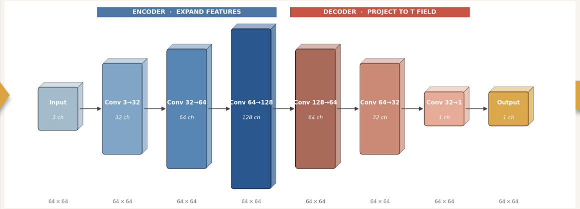
Pfromm, Kanani, Sharma, Solanki, Tervo, Park, Doppa, Pande, Ogras. MFIT: Multi-Fidelity Thermal Modeling for 2.5D and 3D Multi-Chiplet Architectures. arXiv:2410.09188 (2024).

Solution: builder GUI with AI surrogate trained on simulations

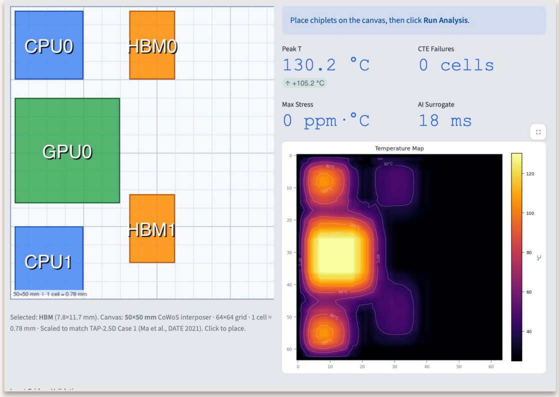
Physics Engine *data_generator.py*



ML surrogate *model_trainer.py*

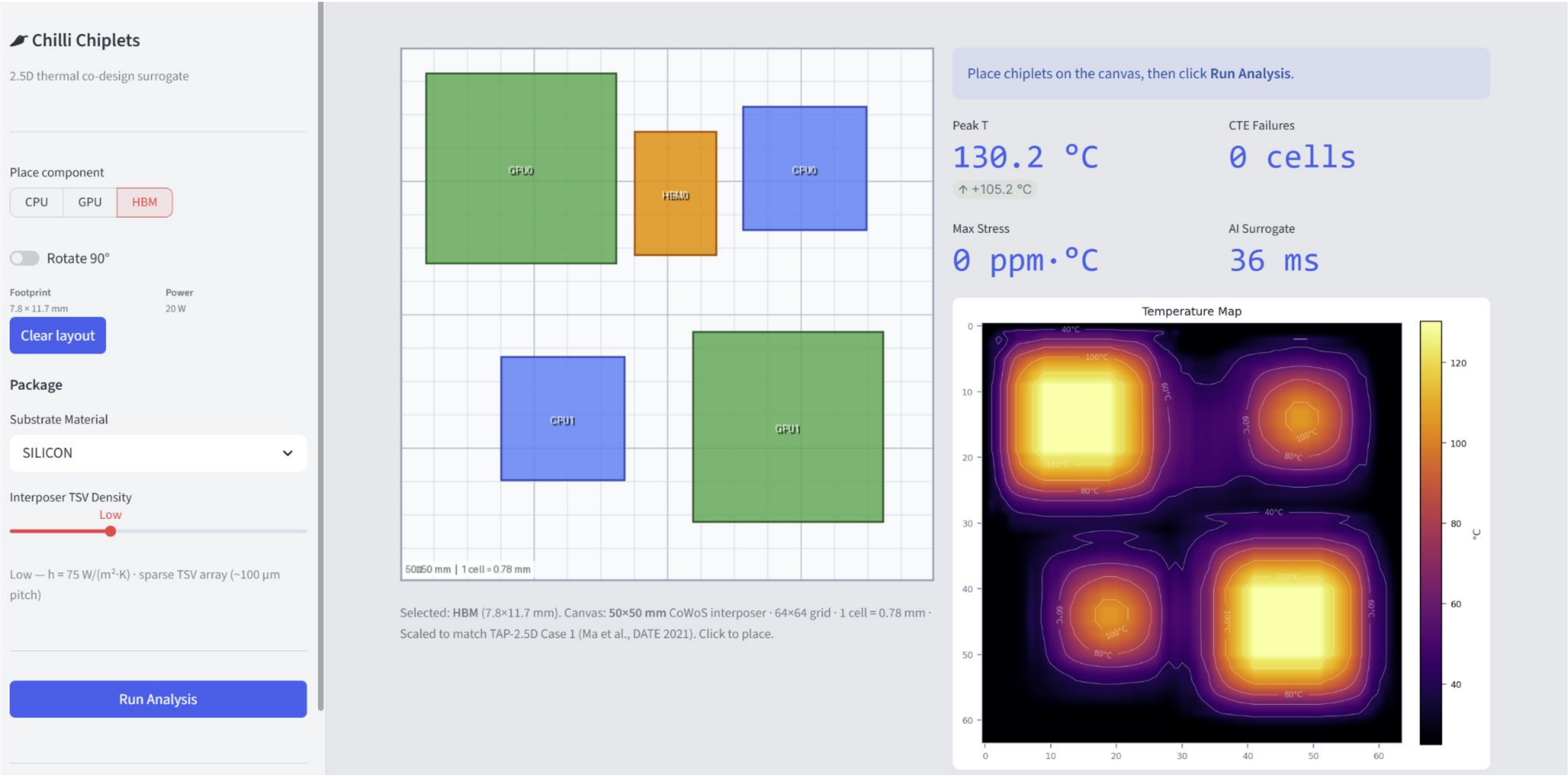


Engineer UI *app.py (Streamlit)*



layout design+thermo+copilot query

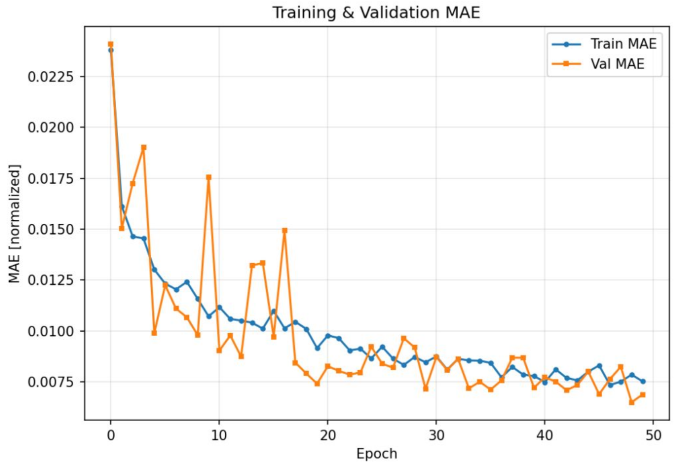
Demo time



Chilli Chiplets
2.5D thermal co-design surrogate
Place component
CPU
GPU
HBM
Rotate 90°
Footprint
7.8 × 11.7 mm
Power
20 W
Clear layout
Package
Substrate Material
SILICON
Interposer TSV Density
Low
Low — h = 75 W/(m²·K) · sparse TSV array (~100 µm pitch)
Run Analysis
1 cell = 0.78 mm
Selected: HBM (7.8×11.7 mm). Canvas: 50×50 mm CoWoS interposer · 64×64 grid · 1 cell = 0.78 mm · Scaled to match TAP-2.5D Case 1 (Ma et al., DATE 2021). Click to place.
Place chiplets on the canvas, then click Run Analysis.
Peak T
130.2 °C
↑ +105.2 °C
CTE Failures
0 cells
Max Stress
0 ppm·°C
AI Surrogate
36 ms
Temperature Map
°C

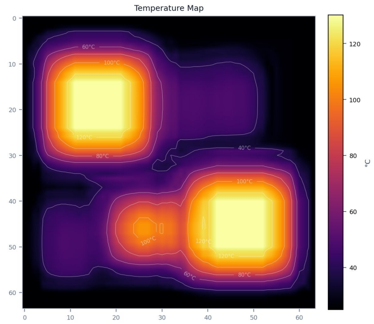
BENCHMARKING

Heat map validation

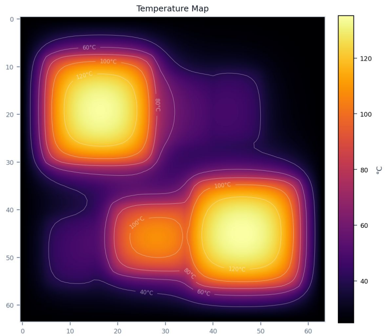


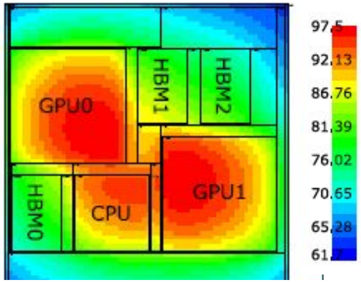
Our AI surrogate model generalized our synthetic Data (2.2 C av. error on 325 C span).
ML MODEL WORKS!!!

AI surrogate
Peak T = 130 C



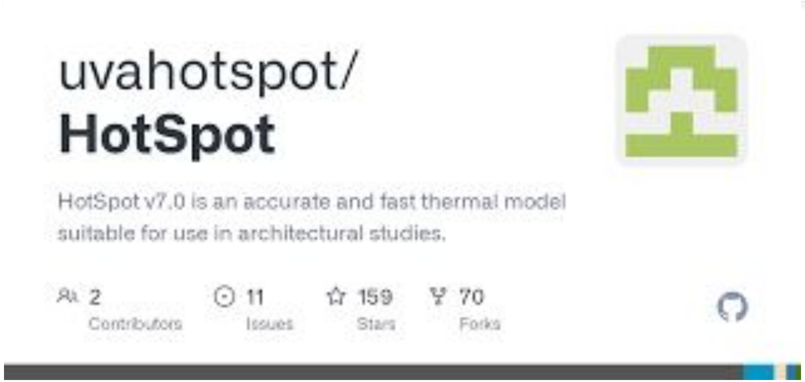
Physics engine
Peak T = 135.3 C

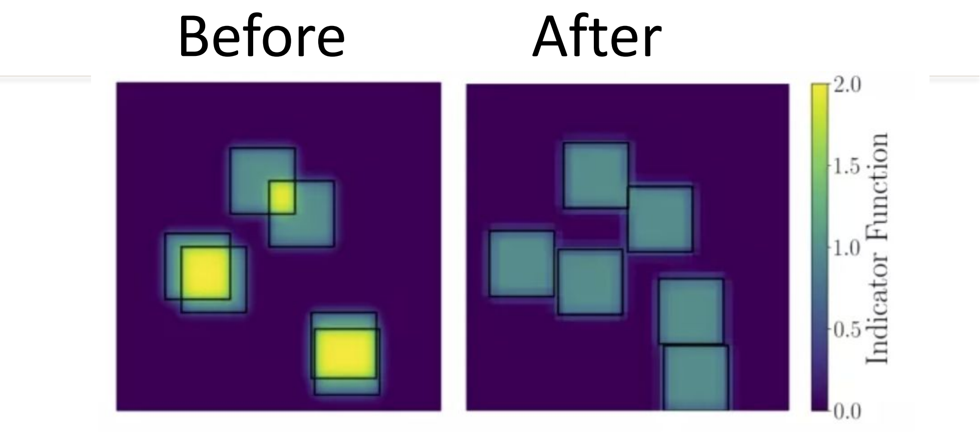




HotSpot simulation
(Ma, et al, 2021)
Peak T = 95.31 C

Next Step?





1. Replace our simple FDM with **HotSpot** as the data factory
    a. production-grade fidelity, same pipeline
2. Add a layout optimizer algorithm
    a. Optimize layout based on chiplet **thermal issues** and **communication latency** and → **conflicting hardware constraints**